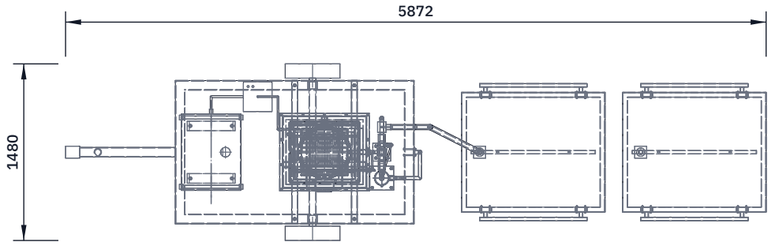
CONCEPT, NOT FOR FABRICATION

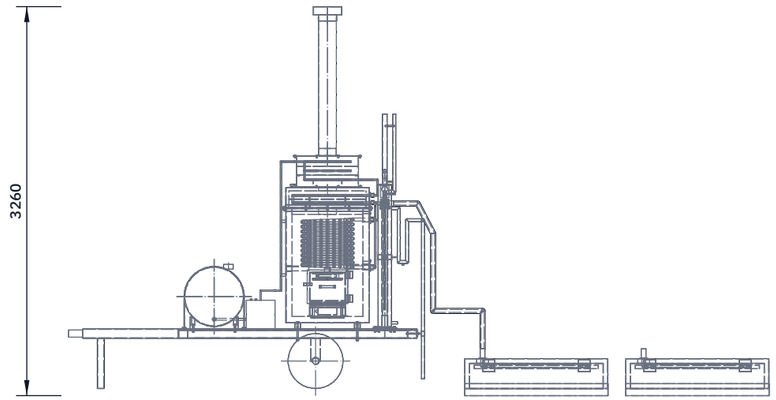
SteamRoot · github.com/BoujeeEnjinia1701/steamroot

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA from TRL 3 model; regulatory ruling open	2026-09-25	AC
P2	Notes per STR-DDR-002: relief valve capacity, towed mass, R6 limits	2026-09-25	AC
P3	Constructable design per STR-DDR-003; making sketches STR-DWG-101 to 121	2026-10-01	AC
P4	Evaporator bank, dampers, seal pot overflow, ballast handles (2026-10-02)	2026-10-02	AC
P5	Kit 1.9 D3 model; gas-pass baffles, flue openings and chimney moved (STR-DEC-001 S1)	2026-10-24	AC



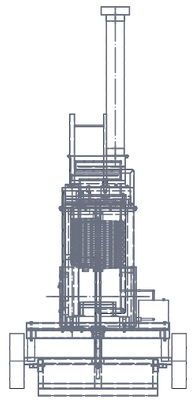
TOP VIEW

Scale 1:50; looking down (along -Z)



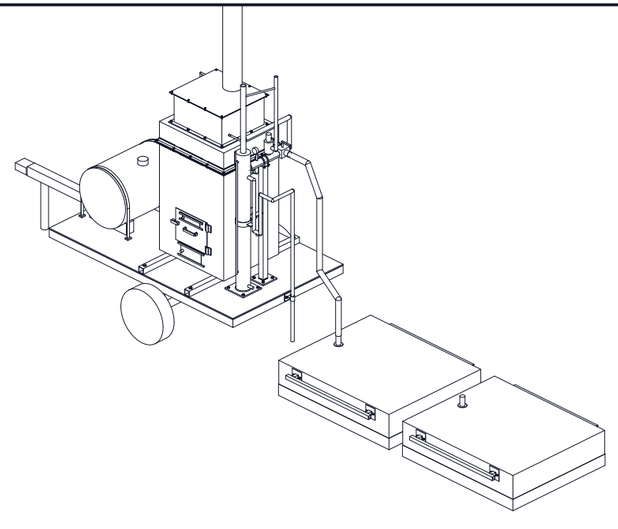
FRONT VIEW

Scale 1:50; looking at the front (along +Y)



RIGHT VIEW

Scale 1:50; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND NOTES

Overall, working position: 5.87 m long, 1.48 m wide, 3.26 m high (mm on sheet)
Deck 2000 x 1200 at 500 above ground; towed 509 kg drained (643 kg full)
Normal pressure: header 0.033 bar (limit 0.1), bank inlet 0.13 bar (limit 0.15); seal 0.084 bar
Steam side always open to the vent; diverter has no closed position; no isolating valve
Hose moved between hoods only after steam is diverted to the vent
Water in heated section 7.3 L flooded; 30 kg/h steam, 33 kW firing (CAL-001)
Firebox and chimney outdoors only; generator never inside a greenhouse
OPEN PREREQUISITE: written ruling from the local boiler authority
on this open-vented fired coil. No detail design or build before it.
Making sketches STR-DWG-101 to 128; build plan STR-BLD-001

ITEMS (BOM NUMBERS)

1 Trailer frame and drawbar; 2 Wheels (2)
3 Feed tank 125 L; 4 Feed pump 12 V
5 Firebox 700 x 600 x 960 on skids, 50 fiber; 2 air dampers
6 Coil 316 SS 25.4 x 1.65, 13.2 m, 10 turns, above the fire
22 Evaporator bank 700 x 600 x 170, 6.4 m of 15.88 tube, 1 baffle
7 Economizer 520 x 460 x 260, 7.3 m tube, 2 baffles
8 Chimney DN150 at the back, outlet 3.2 m, 6 mm mesh
9 Header DN50 at 1.55 m; dip leg 890; vent 2.3 m
Seal pot overflow at the water mark, 350 loop seal, to ground at rear
10 Certified relief valve, 15 psi (1.03 bar), rated 170 kg/h
11 Steam hose, 25 bore x 6 m
12 Hoods (2), 1200 x 1000 x 250, handles for 40 kg ballast
17 to 21 Skids, post, stay, saddles; brackets, stand, glands;
feed lines; vent line and relief discharge; fixings

STEAMROOT

github.com/BoujeeEnjinia1701/steamroot

General arrangement, working position

DRAWING NO.	REV	SHEET	SIZE
STR-DWG-002	P5	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
1:50	mm		2026-10-24	Amish Chadha

MATERIAL / NOTES

PRELIMINARY, NOT FOR FABRICATION. Boiler authority ruling OPEN (prerequisite)

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